Practical 9: Time and Space Complexity Analysis (CO2)

Code:

```
import time

def linear_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target:
            return i
    return -1

def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1

data = list(range(100000))
target = 99999

start = time.perf_counter()
linear_search(data, target)
t1 = time.perf_counter() - start

start = time.perf_counter()
binary_search(data, target)
t2 = time.perf_counter() - start

print(f"Linear Search Time: {t1:.6f} s  (Complexity: O(n))")
print(f"Binary Search Time: {t2:.6f} s  (Complexity: O(log n))")
print("Space Complexity: O(1) for both (in-place)")
```

Output:

```
Linear Search Time: 0.002474 s  (Complexity: O(n))
Binary Search Time: 0.000005 s  (Complexity: O(log n))
Space Complexity: O(1) for both (in-place)
```